

No Welfare Ties Between Pure-Strategy Nash Equilibria

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Abstract

We show that distinct pure-strategy Nash equilibria typically have different utilitarian welfare values in sufficiently rich classes of C^2 games with open strategy sets. Here, typical means both generic and finitely prevalent, while rich means that the class admits a finite-dimensional family of perturbations that can independently move payoff values and first derivatives in any direction at any two distinct strategy profiles—for example, polynomial perturbations of degree at most three. Consequently, a welfare-maximizing equilibrium, if one exists, is unique. The result extends to each fixed C^1 payoff aggregator whose derivative with respect to the payoff vector is everywhere nonzero. In the full C^2 utility space, we also treat scalar criteria depending on strategies, payoffs, and payoff derivatives, subject to a nonvanishing condition in payoff-level or cross-player derivative directions. Finally, adjoining arbitrarily small perturbations from a suitable finite-dimensional space makes welfare separation typical in the augmented family, with separation holding for almost every perturbation of each fixed base game.

1 Intuition

Consider a game with n players, each having an open strategy set $X \subseteq \mathbb{R}^d$, so the strategy-profile space is $X^n \subseteq \mathbb{R}^{nd}$. The first-order conditions define a map $F(u, \cdot) : X^n \rightarrow \mathbb{R}^{nd}$, whose zero set contains the equilibria. For pairs of distinct profiles, write

$$\text{Conf}_2(X^n) := \{(x, y) \in X^n \times X^n : x \neq y\}.$$

For a scalar criterion A that is C^1 in profiles, pairs of distinct equilibria with equal A -values are zeros of

$$(x, y) \mapsto (F(u, x), F(u, y), A(u, x) - A(u, y)), \quad \text{Conf}_2(X^n) \rightarrow \mathbb{R}^{2nd+1}.$$

If this map is transverse to $\{0\}$, it has no zero. By the preimage theorem, a nonempty transverse preimage is a submanifold whose dimension is the domain dimension minus the codimension of the target submanifold in the codomain. Here a nonempty zero set would therefore have dimension

$$\dim \text{Conf}_2(X^n) - \text{codim}_{\mathbb{R}^{2nd+1}} \{0\} = 2nd - (2nd + 1) = -1,$$

which is impossible.

Hence, what we will show is that transversality is typical. Starting with utilitarian welfare, we establish this transversality for typical games by applying Hunt, Sauer, and Yorke's parametric transversality theorem [5, Theorem 1].

To the best of our knowledge, the no-welfare-ties result established here is new. Related genericity and prevalence methods, however, have an established history in economic theory. For almost all finite normal-form games, equilibria are regular and persist locally and smoothly under payoff

perturbations [6]; in finite-state, finite-action discounted stochastic games, stationary Markov perfect equilibria are generically purifiable by vanishing private payoff shocks [2]. For C^2 games with at least two players and compact convex polyhedral strategy sets, finiteness of the pure-equilibrium set is finitely prevalent [1]. On an open dense set of C^2 games with at least two players and simplex strategy sets, equilibria at which every player chooses a nonvertex strategy are Pareto-inefficient [3]. With observable preferences and a pure-equilibrium selection that is C^1 at zero types, the joint space of regular C^3 two-player games on open strategy sets with pure equilibria and C^3 disposition families contains relatively prevalent sets where some preference distortions improve actual payoffs and regular payoff-monotonic selection from full-support initial type distributions does not jointly converge to undistorted payoff maximization [4]. The present note establishes a further typical property: distinct pure equilibria have distinct welfare values, without requiring equilibrium finiteness or compact strategy spaces.

2 Setup

Fix integers $n, d \geq 1$. Let $X \subseteq \mathbb{R}^d$ be nonempty and open. The profile space is $X^n \subseteq \mathbb{R}^{nd}$, and the space of C^2 utility profiles is

$$\mathcal{U} := C^2(X^n, \mathbb{R})^n,$$

with the usual C^2 topology.

Throughout, we use the standard Euclidean inner products to identify each finite-dimensional Euclidean space with its dual.

For $u = (u_1, \dots, u_n) \in \mathcal{U}$, write $x = (x_1, \dots, x_n) \in X^n$, where $x_i \in X$ is player i 's strategy and $x_{-i} := (x_j)_{j \neq i}$ is the profile of the other players' strategies. The set of pure-strategy Nash equilibria is

$$\text{NE}(u) := \{x \in X^n : u_i(x_i, x_{-i}) \geq u_i(a_i, x_{-i}) \text{ for every } a_i \in X \text{ and every } i\}.$$

Define utilitarian welfare by

$$W : \mathcal{U} \times X^n \rightarrow \mathbb{R}, \quad W(u, x) := \sum_{i=1}^n u_i(x).$$

Utilities are cardinal payoffs on specified scales; the welfare rule is held fixed as payoff functions vary.

Define the first-order-condition map

$$F : \mathcal{U} \times X^n \rightarrow (\mathbb{R}^d)^n \cong \mathbb{R}^{nd}, \quad F(u, x) := (D_i u_i(x))_{i=1}^n,$$

where $D_i u_i(x)$ is the derivative of u_i with respect to player i 's own strategy. Since X is open, every pure-strategy Nash equilibrium is an interior optimum in each player's own strategy and hence

$$x \in \text{NE}(u) \implies F(u, x) = 0. \quad (1)$$

Define the equilibrium-candidate welfare-tie map

$$T : \mathcal{U} \times \text{Conf}_2(X^n) \rightarrow \mathbb{R}^{2nd+1}, \quad T(u, x, y) := (F(u, x), F(u, y), W(u, x) - W(u, y)). \quad (2)$$

For fixed u , the zeros of $T(u, \cdot, \cdot)$ are pairs of distinct profiles that satisfy every player's first-order condition and tie in welfare. Therefore (1) implies

$$\begin{aligned} & \{(x, y) \in \text{Conf}_2(X^n) : x, y \in \text{NE}(u), W(u, x) = W(u, y)\} \\ & \subseteq \{(x, y) \in \text{Conf}_2(X^n) : T(u, x, y) = 0\}. \end{aligned} \quad (3)$$

Section 3 shows that, typically, $T(u, \cdot, \cdot)$ has no zero. By (3), no pair of distinct Nash equilibria then ties in welfare.

3 No welfare ties

3.1 Hermite interpolation and the probe

Let $\mathcal{P}_{\leq 3}(\mathbb{R}^{nd}, \mathbb{R}^n)$ denote the vector space of polynomial maps from $\mathbb{R}^{nd}$ to $\mathbb{R}^n$ of degree at most three. Its key property is two-point Hermite interpolation: its elements can match arbitrarily specified values and first derivatives at any two distinct points. The first jet records exactly these data. For $q \geq 1$ and $z \in \mathbb{R}^{nd}$, let

$$J_z^1(\mathbb{R}^{nd}, \mathbb{R}^q) := \mathbb{R}^q \times \text{Lin}(\mathbb{R}^{nd}, \mathbb{R}^q).$$

If $O \subseteq \mathbb{R}^{nd}$ is open, $f \in C^1(O, \mathbb{R}^q)$, and $z \in O$, define

$$j_z^1 f := (f(z), Df(z)) \in J_z^1(\mathbb{R}^{nd}, \mathbb{R}^q).$$

For utility profiles, abbreviate $J_z^1 := J_z^1(\mathbb{R}^{nd}, \mathbb{R}^n)$.

For distinct $x, y \in \mathbb{R}^{nd}$, write $J_{x,y}^1 := J_x^1 \times J_y^1$ and $j_{x,y}^1 f := (j_x^1 f, j_y^1 f)$ whenever f is defined on an open set containing x and y .

Let P be the restriction of $\mathcal{P}_{\leq 3}(\mathbb{R}^{nd}, \mathbb{R}^n)$ to X^n .

Lemma 1 (Two-point Hermite interpolation). *Fix distinct $x, y \in X^n$. Then the two-point jet evaluation map*

$$j_{x,y}^1 : P \longrightarrow J_{x,y}^1$$

is surjective and hence a submersion.

Proof. Because the interpolation problem is componentwise, it suffices to treat the scalar-valued case. Fix an arbitrary element $((a, g), (b, h)) \in J_x^1(\mathbb{R}^{nd}, \mathbb{R}) \times J_y^1(\mathbb{R}^{nd}, \mathbb{R})$ of the codomain, where $a, b \in \mathbb{R}$ and $g, h \in (\mathbb{R}^{nd})^*$. Choose an affine linear function $t : \mathbb{R}^{nd} \rightarrow \mathbb{R}$ such that $t(x) = 0$ and $t(y) = 1$. Define

$$H_0(s) := 2s^3 - 3s^2 + 1, \quad H_1(s) := -2s^3 + 3s^2,$$

and set

$$q(z) := aH_0(t(z)) + bH_1(t(z)) + (1 - t(z))^2 g(z - x) + t(z)^2 h(z - y).$$

Then q has degree at most three and satisfies

$$q(x) = a, \quad Dq(x) = g, \quad q(y) = b, \quad Dq(y) = h.$$

Applying this construction independently to each component and restricting the resulting polynomial map to X^n proves the claim. $\square$

We use the following probe formulation: a set $A \subseteq \mathcal{U}$ is *finitely prevalent with probe P* if it contains a Borel set $B \subseteq \mathcal{U}$ such that, for every $u \in \mathcal{U}$,

$$u + p \in B \quad \text{for Lebesgue almost every } p \in P.$$

A set is *generic* if it contains a countable intersection of open dense sets, and *typical* if it is both generic and finitely prevalent.

3.2 Application to welfare ties

We use the following corollary of Hunt, Sauer, and Yorke's parametric transversality theorem [5, Theorem 1], specialized to the target submanifold $\{0\}$ and applied in local coordinates.

Corollary 2 (Parametric avoidance of zero). *Let A and B be open subsets of finite-dimensional, second-countable smooth manifolds without boundary, and let $\Phi : A \times B \rightarrow \mathbb{R}^q$ be C^1 , with $q > \dim A$. If Φ is transverse to $\{0\}$, then for almost every $b \in B$,*

$$\Phi(a, b) \neq 0 \quad \text{for every } a \in A.$$

Here almost every means outside a set of Lebesgue measure zero in local coordinates on B .

Theorem 3 (No utilitarian welfare ties). *The set*

$$\left\{ u \in \mathcal{U} : \text{for all } x, y \in \text{NE}(u), x \neq y \implies \sum_{i=1}^n u_i(x) \neq \sum_{i=1}^n u_i(y) \right\}$$

is typical in $\mathcal{U}$, with probe P .

Proof. Let

$$\mathcal{G} := \{u \in \mathcal{U} : T(u, x, y) \neq 0 \text{ for every } (x, y) \in \text{Conf}_2(X^n)\}.$$

Fix $u \in \mathcal{U}$ and define the C^1 map

$$\Phi_u : \text{Conf}_2(X^n) \times P \rightarrow \mathbb{R}^{2nd+1}, \quad \Phi_u(x, y, p) := T(u + p, x, y).$$

For each $x \neq y$, Lemma 1 lets a perturbation $h \in P$ prescribe independently all utility values and first derivatives at x and y . In particular, it can prescribe both FOC blocks and the welfare difference independently. Since T is linear in utilities, $D_p \Phi_u(x, y, p)$ is the surjective map $h \mapsto T(h, x, y)$. Thus Φ_u is a submersion.

As $\dim \text{Conf}_2(X^n) = 2nd < 2nd + 1$, Corollary 2 gives $u + p \in \mathcal{G}$ for almost every $p \in P$, for every $u \in \mathcal{U}$. Such perturbations exist arbitrarily close to zero; continuity of $p \mapsto u + p$ therefore also makes $\mathcal{G}$ dense.

The open set $\text{Conf}_2(X^n)$ admits a countable compact exhaustion. Nonvanishing of T on each compact set is open in u , by uniform control of payoff values and first derivatives on its coordinate projections. Thus $\mathcal{G}$ is a dense G_δ , hence generic and Borel. The same almost-everywhere conclusion establishes its finite prevalence with probe P . By (3), every game in $\mathcal{G}$ has no distinct welfare-tied Nash equilibria, proving the claim. $\square$

Corollary 4 (Unique equilibrium maximizing utilitarian welfare). *For every game in the typical set of Theorem 3, a pure-strategy Nash equilibrium maximizing utilitarian welfare, if one exists, is unique.*

Proof. Two distinct equilibria maximizing utilitarian welfare would have the same utilitarian welfare, contrary to Theorem 3. $\square$

Remark 5 (Minimal regularity for this argument). *The first-order-condition map F uses one derivative of each utility, and parametric transversality requires $T(u, \cdot, \cdot)$ to be at least C^1 . Thus $u_i \in C^2$ is the weakest standard regularity under which this proof applies.*

Remark 6 (Scope). *The theorem as stated concerns pure equilibria on an open strategy set and utilitarian welfare; it does not assert equilibrium existence or treat mixed strategies. Section 4 records extensions of the welfare criterion, probe, and utility space.*

4 Generalizations

4.1 Payoff aggregators

Proposition 7 (No ties for payoff aggregators). *Fix $H \in C^1(X^n \times \mathbb{R}^n, \mathbb{R})$ with $D_v H(z, v) \neq 0$ for every (z, v) , and define*

$$A_H(u, z) := H(z, u(z)).$$

The set of games whose distinct pure-strategy Nash equilibria have distinct values of A_H is typical in $\mathcal{U}$, with probe P .

The gradient condition means that, holding strategies fixed, changing at least one player's payoff changes the index to first order. The typical set may depend on the fixed aggregator H .

Proof. Replace T by

$$T_H(u, x, y) := (F(u, x), F(u, y), A_H(u, x) - A_H(u, y)).$$

For each fixed u , the map $(x, y, p) \mapsto T_H(u + p, x, y)$ is C^1 . Lemma 1 lets a perturbation $h \in P$ prescribe both FOC blocks and the values $h(x), h(y)$ independently. Setting $h(y) = 0$, the payoff-value contribution to the differential of the last coordinate is

$$D_v H(x, (u + p)(x))h(x).$$

This can take any scalar value without altering either FOC block. Hence the parameter derivative is surjective. Corollary 2 gives absence of zeros of $T_H(u + p, \cdot, \cdot)$ for almost every $p \in P$, for every base game u .

As in the proof of Theorem 3, nonvanishing on each compact subset of $\text{Conf}_2(X^n)$ is open in the C^2 topology. Compact exhaustion and arbitrarily small good perturbations therefore make the set of games with no zeros of T_H a dense G_δ . It is the Borel witness for finite prevalence, and its games have no distinct A_H -tied Nash equilibria. $\square$

Corollary 8 (Weighted welfare). *For each fixed $\lambda \in [0, \infty)^n \setminus \{0\}$, distinct pure-strategy Nash equilibria typically have distinct values of $W_\lambda(u, x) := \sum_i \lambda_i u_i(x)$.*

Proof. Take $H(z, v) = \sum_i \lambda_i v_i$, so $D_v H = \lambda \neq 0$. $\square$

Corollary 9 (Unique equilibrium maximizing a payoff aggregator). *For each fixed H as in Proposition 7 and every game in its typical set, a pure-strategy Nash equilibrium maximizing A_H , if one exists, is unique.*

Proof. Two distinct maximizing equilibria would tie in A_H , contrary to Proposition 7. $\square$

Appendix A extends the separation result to scalar criteria that also depend on payoff derivatives.

4.2 Alternative probes

A finite-dimensional linear subspace $P_0 \subseteq \mathcal{U}$ is a *probe for $\mathcal{U}$* if, for every $u \in \mathcal{U}$, the map $(p, x, y) \mapsto j_{x,y}^1(u + p)$ is C^1 on $P_0 \times \text{Conf}_2(X^n)$ and its derivative in p is surjective everywhere. Addition uses the chosen linear structure on $\mathcal{U}$. Under the ordinary pointwise operations, joint C^1 regularity follows from the C^2 regularity of the utilities, and

$$j_{x,y}^1(u + p) = j_{x,y}^1 u + j_{x,y}^1 p,$$

so the probe condition reduces to surjectivity of $j_{x,y}^1|_{P_0}$. By Lemma 1, the cubic space P is a probe for every pointwise-linear utility space containing it. Any finite-dimensional subspace containing a probe is also a probe in this pointwise setting.

4.3 Admissible utility spaces

Definition 10 (Admissible utility space). *A class $\mathcal{U} \subseteq C^2(X^n, \mathbb{R})^n$ is admissible if it is endowed with a topological vector-space structure for which the evaluation map*

$$\text{ev} : \mathcal{U} \times X^n \longrightarrow \mathbb{R}^n, \quad \text{ev}(u, z) := u(z),$$

is jointly continuous. The linear operations need not be pointwise.

The space $C^2(X^n, \mathbb{R})^n$, with its usual C^2 topology, is admissible, as is every pointwise-linear subspace with the induced topology. This includes $\mathcal{P}_{\leq k}(\mathbb{R}^{nd}, \mathbb{R}^n)|_{X^n}$ for every $k \geq 3$.

The conclusions of Theorem 3 and Proposition 7 hold typically in any admissible utility space admitting a probe, for each fixed H . The probe condition and the proposition's rank argument give a C^1 submersion along each probe translate, hence almost-everywhere absence of ties. Joint evaluation continuity and continuity of H make the relation of A_H -tied Nash pairs closed, since every Nash inequality and aggregator equality is a closed condition. Its projection over each compact subset of $\text{Conf}_2(X^n)$ is closed, so compact exhaustion makes the Nash no-ties set G_δ . Small good probe perturbations make this set dense; it is also the Borel witness for finite prevalence.

5 Arbitrarily small augmentation

Return to utilitarian welfare on the open strategy space X . Even when welfare separation is not typical in an admissible utility class, adjoining arbitrarily small payoff perturbations from a suitable probe makes separation generic in the augmented family and full measure along every base-game fiber.

Let $\mathcal{V}$ be an admissible utility space as in Section 4.3, and let P_0 be any probe for $C^2(X^n, \mathbb{R})^n$ with its pointwise linear operations. Thus P_0 is a finite-dimensional pointwise-linear space of payoff perturbations with $j_{x,y}^1|_{P_0}$ surjective for every $x \neq y$. Form the external direct sum, with the product topology,

$$\tilde{\mathcal{V}} := \mathcal{V} \oplus P_0 = \mathcal{V} \times P_0, \quad \tilde{\text{ev}}((u, p), z) := u(z) + p(z).$$

The embedding $u \mapsto (u, 0)$ preserves the original linear structure. Pairs are retained as parameters, even if they represent the same payoff profile; genericity and prevalence refer to this parameter space.

Corollary 11 (Small augmentation). *Welfare separation is typical in $\tilde{\mathcal{V}}$, with probe $\{0\} \times P_0$. Fix any norm on P_0 . For each $u \in \mathcal{V}$ and $\varepsilon > 0$, a generic, full-Lebesgue-measure subset of*

$\{p \in P_0 : \|p\| < \varepsilon\}$ yields no welfare ties between distinct Nash equilibria. For every compact $S \subset X^n$,

$$\max_{|\alpha| \leq 2} \sup_{z \in S} \|D^\alpha p(z)\| \leq C_{S, P_0} \|p\|,$$

with $C_{S, P_0} > 0$ independent of u and p . The bound also holds with $S = X^n$ if all elements of P_0 have bounded derivatives of orders zero through two on X^n .

Proof. Translation by $(0, q)$ adds the payoff q , so the probe condition and the proof of Theorem 3 give full measure along every probe translate. Joint evaluation continuity makes the relation of welfare-tied Nash pairs closed. Compact projection and exhaustion of $\text{Conf}_2(X^n)$ therefore make the no-ties set G_δ ; small good probe perturbations make it dense. This also supplies the Borel witness for finite prevalence. On each fixed-base probe ball, the good set is again G_δ and has full measure, hence is generic. The bound follows by expanding p in a basis of P_0 and bounding the basis functions and their derivatives on S , or on all of X^n under the stated boundedness condition. $\square$

For each fixed H as in Proposition 7, Corollary 11 also holds with welfare replaced by A_H , evaluated at augmented payoffs as $H(z, u(z) + p(z))$. The proposition supplies the same rank argument along $\{0\} \times P_0$, and continuity of H preserves the closed relation of tied Nash pairs.

For example, the cubic polynomial space P is such a probe by Lemma 1. When X is precompact, its polynomial derivatives are bounded on $\overline{X^n}$, giving uniform C^2 smallness on the whole domain.

Remark 12 (A globally bounded probe). Set $w(z) := e^{-\|z\|^2}$ and $P_b := wP$. At each z , multiplication by w induces the invertible first-jet map

$$(v, L) \longmapsto (w(z)v, w(z)L + v Dw(z)).$$

Thus Lemma 1 gives two-point first-jet surjectivity for P_b . Its elements are restrictions of Gaussian-weighted polynomials, whose derivatives of orders zero through two are bounded on $\mathbb{R}^{nd}$. Corollary 11 therefore gives uniform C^2 -smallness on the whole domain even when X is unbounded.

Thus $\mathcal{V} \times \{p \in P_0 : \|p\| < \varepsilon\}$ is an arbitrarily small augmentation with generic welfare separation, retaining the original class at $p = 0$. Smallness holds through second derivatives on compact subsets, and uniformly under the boundedness condition above. Preserving additional economic restrictions requires perturbation directions that respect them.

A Criteria depending on payoff derivatives

In the full utility space $\mathcal{U} = C^2(X^n, \mathbb{R})^n$, with its usual C^2 topology, the same argument separates scalar criteria involving marginal effects. For $L \in \text{Lin}(\mathbb{R}^{nd}, \mathbb{R}^n)$, write L_{ij} for the derivative block of output i with respect to input block j . When $L = Du(z)$, these blocks are $L_{ij} = D_j u_i(z)$.

Theorem 13 (No ties for nonflat scalar first-jet criteria). Fix a C^1 function

$$\gamma : X^n \times \mathbb{R}^n \times \text{Lin}(\mathbb{R}^{nd}, \mathbb{R}^n) \rightarrow \mathbb{R}, \quad A(u, z) := \gamma(z, u(z), Du(z)).$$

Suppose that

$$D_{(v, (L_{ij})_{i \neq j})} \gamma(z, v, L) \neq 0 \quad \text{for every } (z, v, L).$$

Then the set of games whose distinct pure-strategy Nash equilibria have distinct values of A is typical in $\mathcal{U}$, with probe P .

The condition requires a first-order response to payoff levels or cross effects while holding the own-strategy derivatives fixed. For example, it holds for $A(u, z) = D_j u_i(z)$ when $d = 1$ and $i \neq j$. The typical set may depend on the fixed criterion A .

Proof. Define

$$T_A(u, x, y) := (F(u, x), F(u, y), A(u, x) - A(u, y)).$$

For each fixed u , the map $(x, y, p) \mapsto T_A(u + p, x, y)$ is C^1 . Lemma 1 lets a perturbation prescribe the diagonal derivative blocks at x and y , hence both FOC blocks. The nonflatness condition lets its payoff values and cross derivatives at x supply any remaining change to the last coordinate of the parameter differential, without altering either FOC block or the jet at y . The parameter derivative is therefore surjective, and Corollary 2 gives absence of zeros for almost every $p \in P$, for every base game u .

As in the proof of Theorem 3, nonvanishing on compact configuration sets is open in the C^2 topology. Compact exhaustion and arbitrarily small good perturbations make the no-zero set a dense G_δ and a Borel witness for finite prevalence. Every game in this set has no distinct A -tied Nash equilibria. $\square$

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